

BIOMEDICAL KNOWLEDGE INTELLIGENCE

Eugene

A proprietary knowledge graph fused with live web & literature —
answers you can see, trust, and trace.

Eugene KG

Web

PubMed

Generic LLMs fall short in biomedicine

- **No proprietary data**
They don't know your internal drugs, targets, trials, patents, or org portfolio
- **They hallucinate links**
Confident but unverifiable drug–gene–disease connections
- **Stale knowledge**
Frozen at a training cutoff — no live trials or this-month's papers
- **No provenance**
You can't see where an answer came from or trace the reasoning

WHAT IT IS

One agent. Three sources. Full provenance.

- **Eugene Knowledge Graph**
A 4M-relationship biomedical graph — drugs, genes, diseases, trials, patents, orgs
- **Live Web**
Real-time ClinicalTrials.gov, Google Patents, and open-web fetch
- **Live PubMed / Europe PMC**
Current biomedical literature with downloadable papers
- **Every answer is source-badged**
Green badge shows Eugene KG / Web / PubMed — and an interactive reasoning graph

THE KNOWLEDGE GRAPH

PrimeKG-scale biomedical graph

129,375

Entities (nodes)

4.05M

Relationships

10

Entity types

31

Relationship types

Entity types: drug · gene/protein · disease · pathway · anatomy · phenotype ·
biological process · molecular function · cellular component · exposure

HOW IT WORKS

Architecture

Next.js UI

Chat + interactive Cytoscape reasoning graph, source tabs

Agent (Strands + LLM)

Plans, routes to the selected source, calls tools, cites sources

MCP tools + live tools

Graph lookups/paths/facts · PubMed · ClinicalTrials · Patents

Neo4j + GDS

4M-edge graph, fulltext-indexed lookups, graph-data-science algorithms

REAL STRENGTHS

What only Eugene can do

- **Multi-hop reasoning**
drug → target protein → disease, chained automatically
- **Explainable paths**
Shortest path between any two entities, visualized
- **Drug intelligence**
Targets, interactions, contraindications, side effects
- **Live evidence**
Recruiting trials & this-year's papers, on demand
- **Provenance built-in**
Source-badged answers + evidence chains
- **See & download**
Reasoning graph + downloadable research papers

EXPERIENCE

See it. Trust it. Trace it.

See it

Interactive reasoning graph — expand, collapse, find paths, 20 nodes at a time

Trust it

Green source badge on every answer:
Eugene KG / Web / PubMed

Trace it

Evidence chains + downloadable papers (Read / Download PDF)

DEMO NARRATIVE

One theme, three sources: "Clozapine"

Eugene KG

What proteins does Clozapine target, and what are its side effects?

Web

List current recruiting clinical trials for clozapine, with phases & sponsors.

PubMed

Find recent papers on clozapine-induced agranulocytosis — with links.

Same question, three lenses — proprietary graph + live web + live literature, each source-badged.

THE BOTTOM LINE

Deep biomedical reasoning, made visible and trustworthy.

Eugene fuses a 4M-relationship knowledge graph with live web & PubMed — answering questions a generic LLM can't, with full, traceable provenance.

Eugene KG

Web

PubMed